

NESMR Plotting

Simulation Example

```
set.seed(13)
library(esmr)
library(GWASBrewer)
G <- matrix(
  c(0, 0, 0, 0,
     0.25, 0, 0, 0,
     0, 0, 0, 0,
     0, -0.15, 0.2, 0),
  nrow = 4,
  byrow = 4
)

d <- ncol(G)

B_full <- matrix(1, d, d)
B_lower <- lower.tri(G) + 0
B_correct <- (G != 0) + 0

h2 <- 0.2
J <- 5000
N <- 20000
pi_J <- 0.1
alpha <- 5e-8

dat <- sim_mv(
  G = G,
  N = N,
  J = J,
  h2 = h2,
```

```

    pi = pi_J,
    sporadic_pleiotropy = TRUE,
    est_s = TRUE
)
##> SNP effects provided for 5000 SNPs and 4 traits.

```

Discovery

```

suppressWarnings(capture.output(mh_results <- mh_graph_explore(dat), file = nullfile()))
##> R is not supplied, fitting assuming full independence
##> R is not supplied, fitting assuming full independence
##> R is not supplied, fitting assuming full independence
##> R is not supplied, fitting assuming full independence

kableExtra::kable(discovery_summary(mh_results), digits = 4, row.names = F)

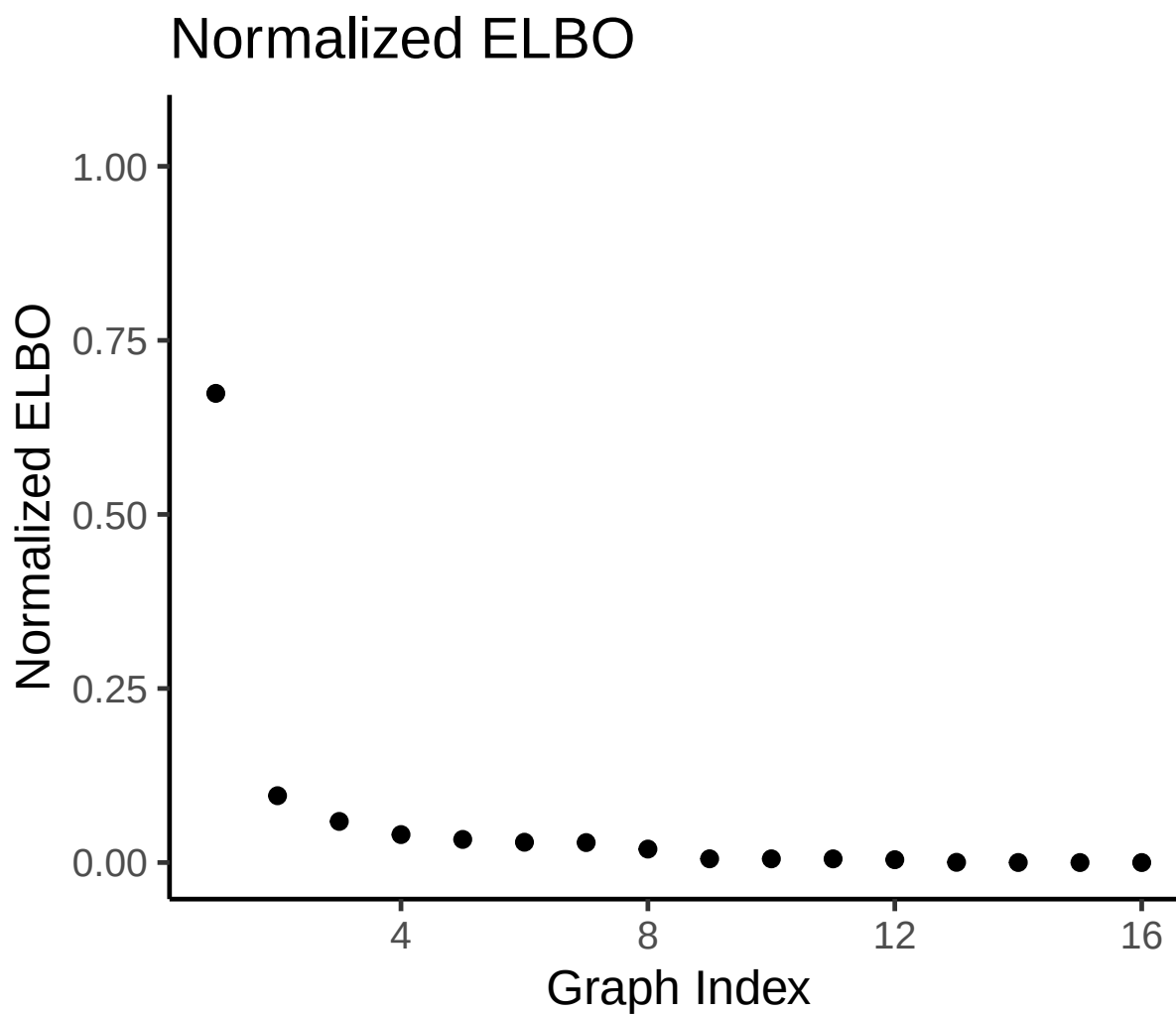
```

graph	num_edges	elbo	norm_elbo	visit_count
0100000100010000	3	1748.117	0.6739	920
0100000000010000	2	1746.168	0.0960	52
0100001100010000	4	1745.682	0.0591	7
0100000101010000	4	1745.297	0.0402	5
0100000110010000	4	1745.107	0.0332	0
0101000100010000	4	1744.977	0.0292	0
0110000100010000	4	1744.960	0.0287	1
0100001000010000	3	1744.566	0.0193	6
0100000001010000	3	1743.287	0.0054	2
0100000000011000	3	1743.280	0.0053	1
0100000000010100	3	1743.269	0.0053	4
0101000000010000	3	1743.011	0.0041	1
0100000001010100	4	1740.475	0.0003	0
0100000100000000	2	1739.134	0.0001	0
0100000000000000	1	1737.380	0.0000	0
0000000100010000	2	1731.552	0.0000	0

Discovery Plots - ELBO

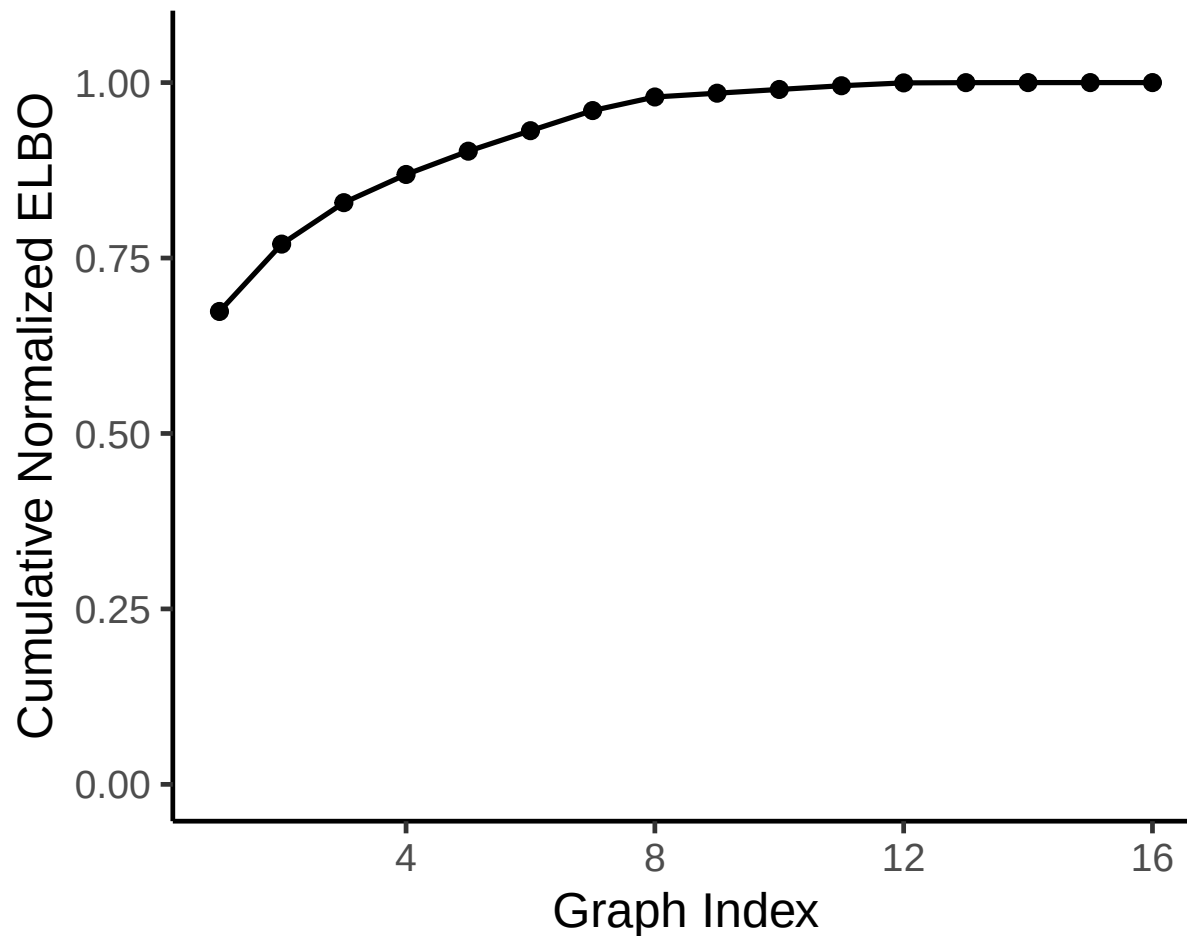
```
library(ggplot2)

plot(mh_results, plot_type = "norm_elbo")
```



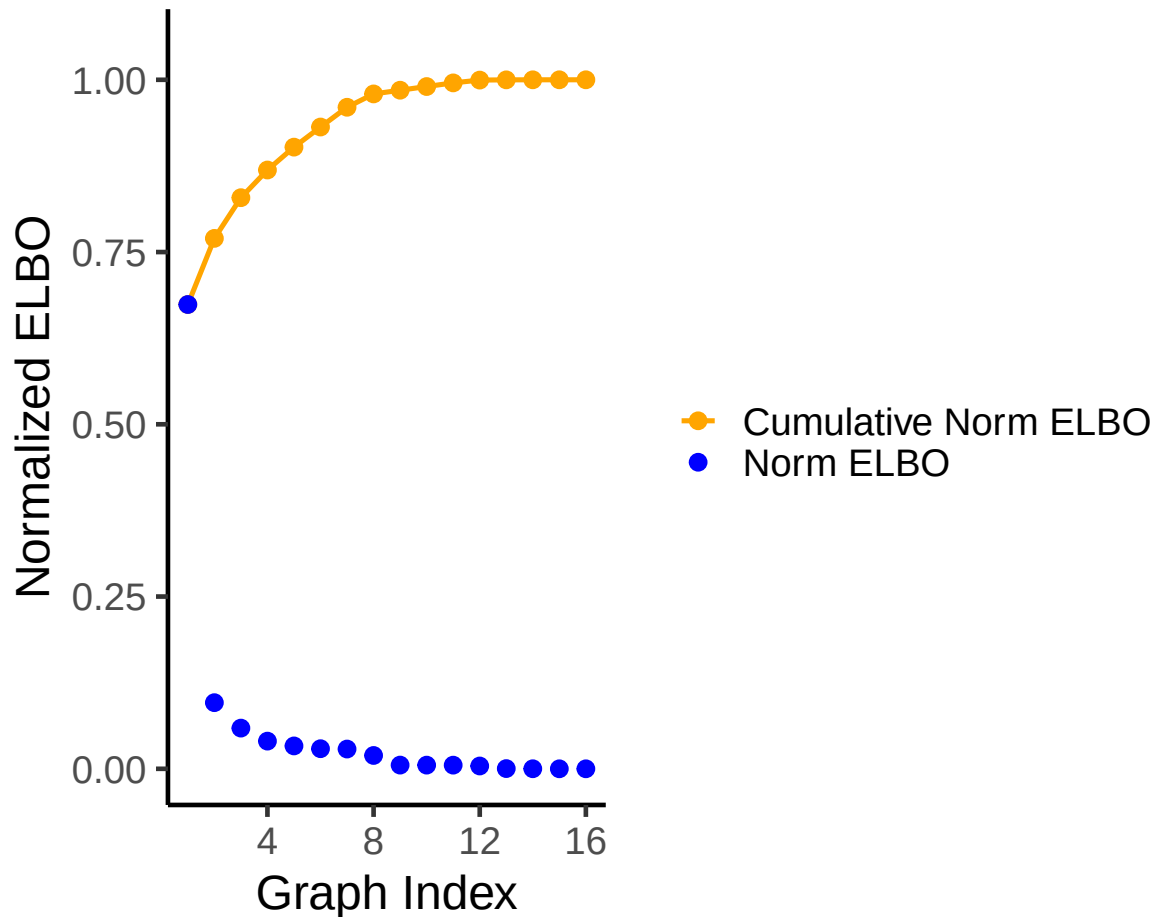
```
plot(mh_results, plot_type = "cum_norm_elbo")
```

Cumulative Normalized ELBO



```
plot(mh_results, plot_type = "both")
```

Normalized ELBO and Cumulative ELBC



Individual Graph Plots

Individual graph objects extend [tidygraphs](#) which provide an easy interface that works with [gggraph](#) for plotting.

We can get the best graph from the discovery results.

```
best_graph <- top_i_graph(mh_results)

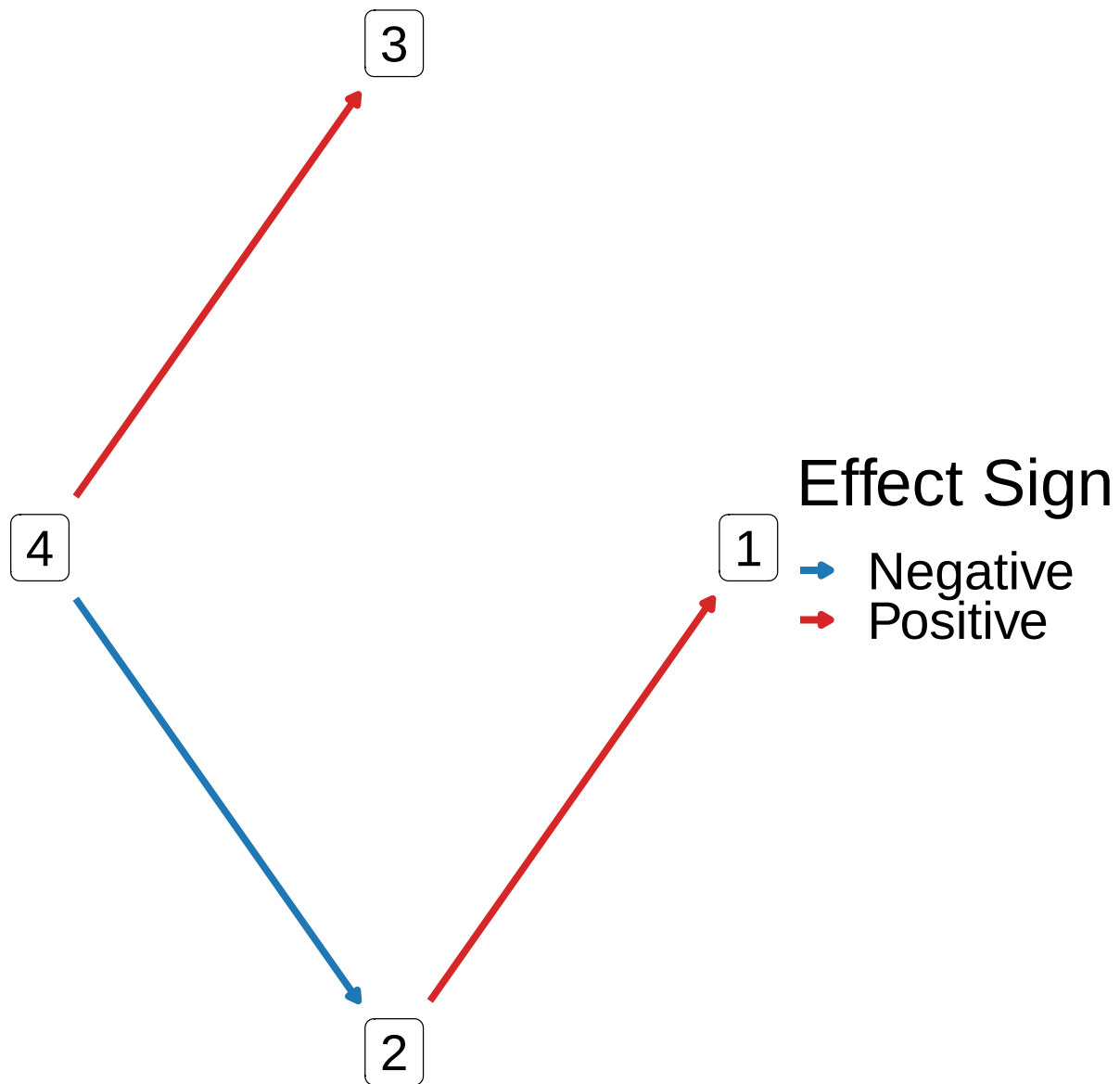
print(best_graph)
##> esmr tidygraph object
##> Evidence lower bound (ELBO) objective function: 1748.117
##> Normalized ELBO (Approximate posterior probability): 0.67
##> # A tbl_graph: 4 nodes and 3 edges
```

```
##> #
##> # A rooted tree
##> #
##> # Node Data: 4 x 1 (active)
##>   name
##>   <int>
##> 1     1
##> 2     2
##> 3     3
##> 4     4
##> #
##> # Edge Data: 3 x 4
##>   from   to direct_effect direct_effect_se
##>   <int> <int>         <dbl>         <dbl>
##> 1     2     1         0.256         0.0256
##> 2     4     2        -0.118         0.0258
##> 3     4     3         0.190         0.0261
```

Adjacency Graph

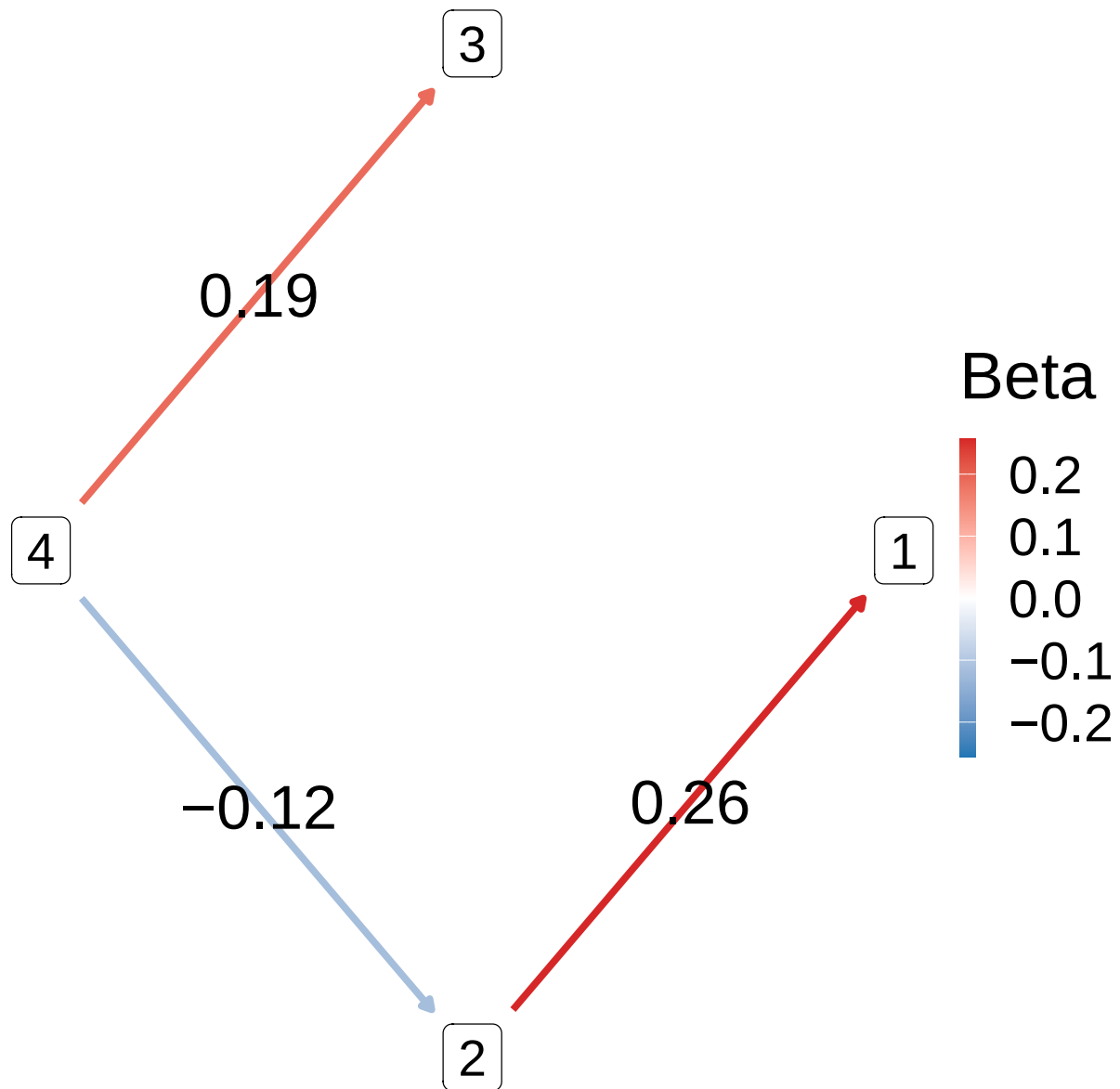
Plot trinary graph (+1,0,-1) for positive and negative effects

```
plot_layered_topo(best_graph)
```



Weighted Graph

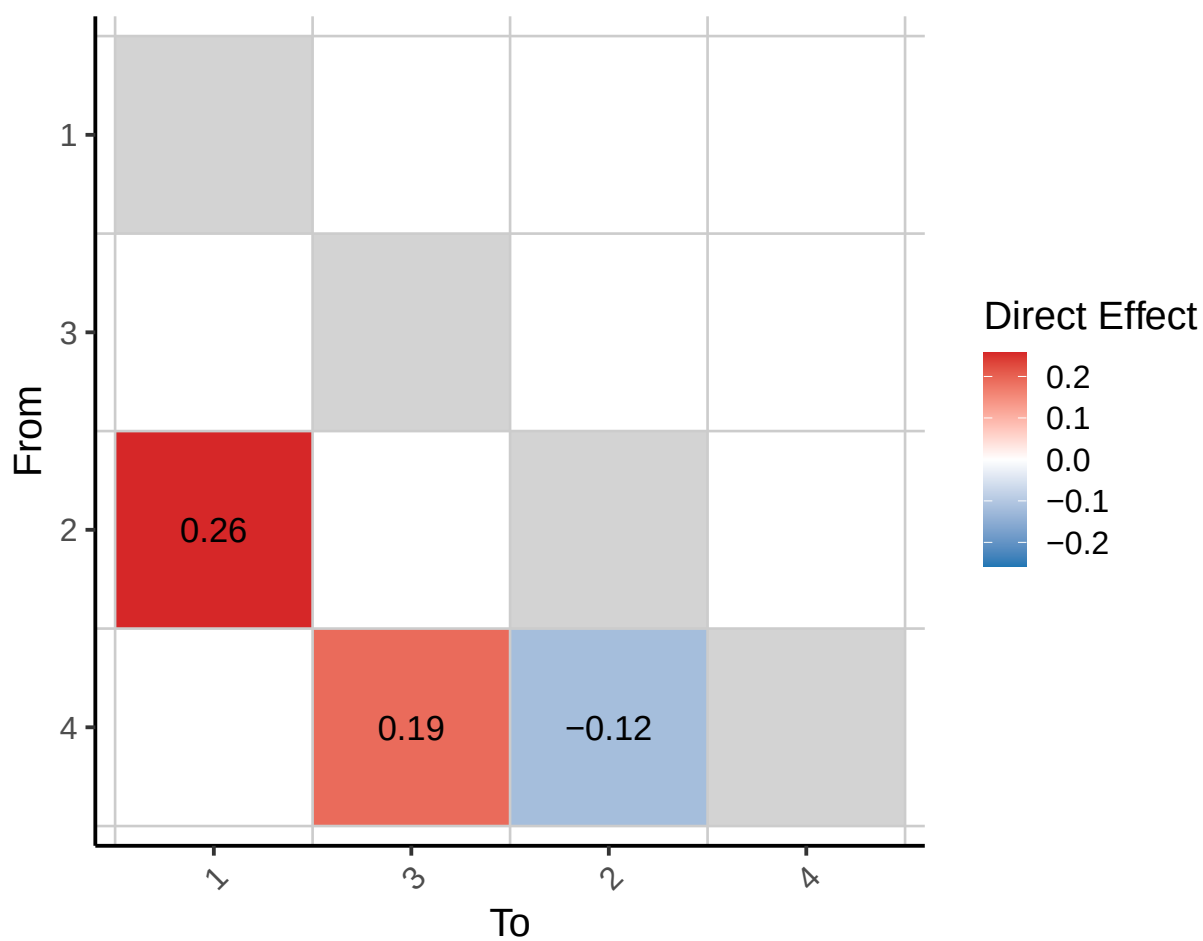
```
plot_layered_topo(best_graph, plot_type = "beta", weight = "direct_effect")
```



Weighted Graph - Tile

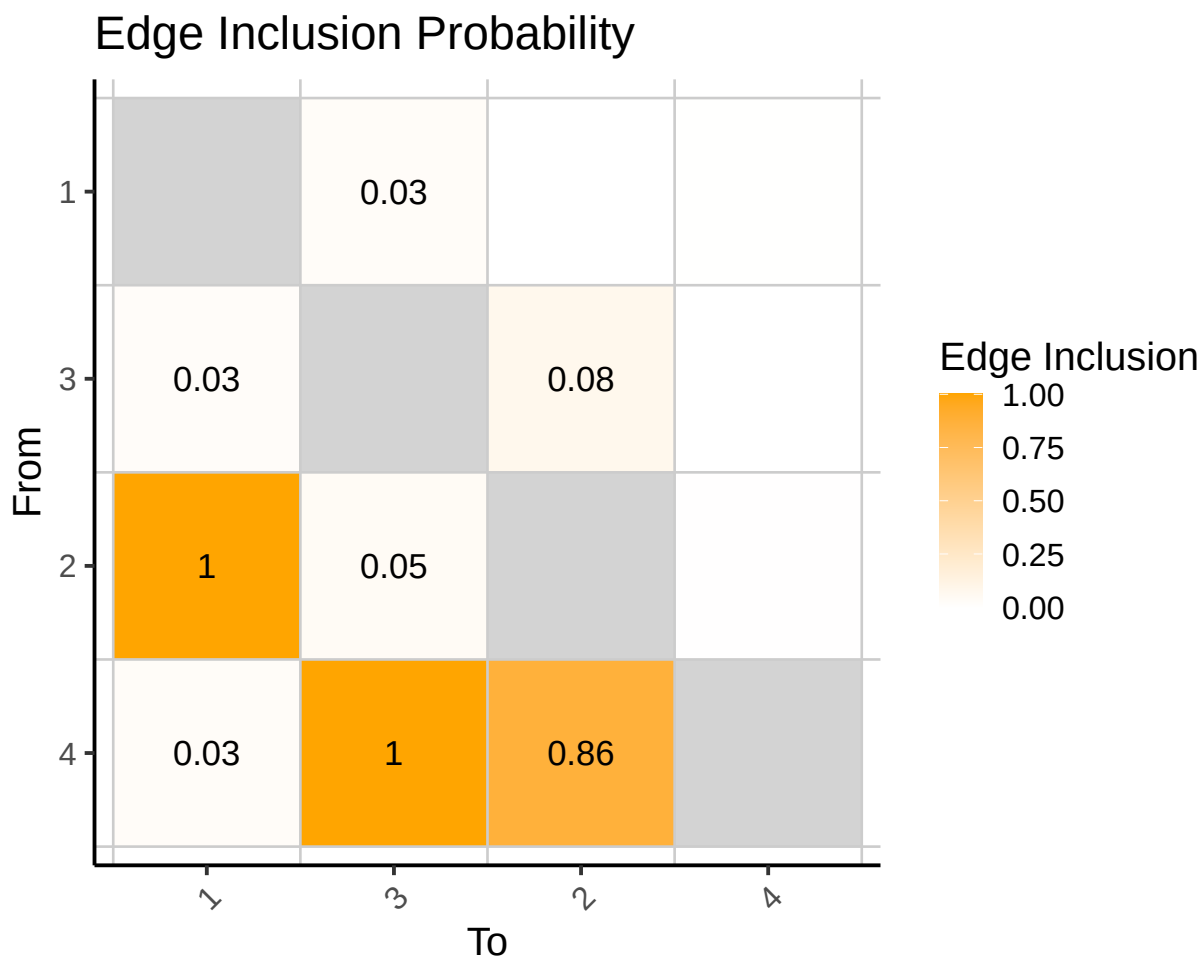
```
plot_nesmr_tile(best_graph)
##> Warning: Removed 4 rows containing missing values or values outside the scale range
##> (`geom_text()`).
```


Direct Effect Matrix



Edge Inclusion Probabilities - Tile

```
edge_inclusion_tile_plot(mh_results)
##> Warning: Removed 4 rows containing missing values or values outside the scale range
##> (`geom_text()`).
```



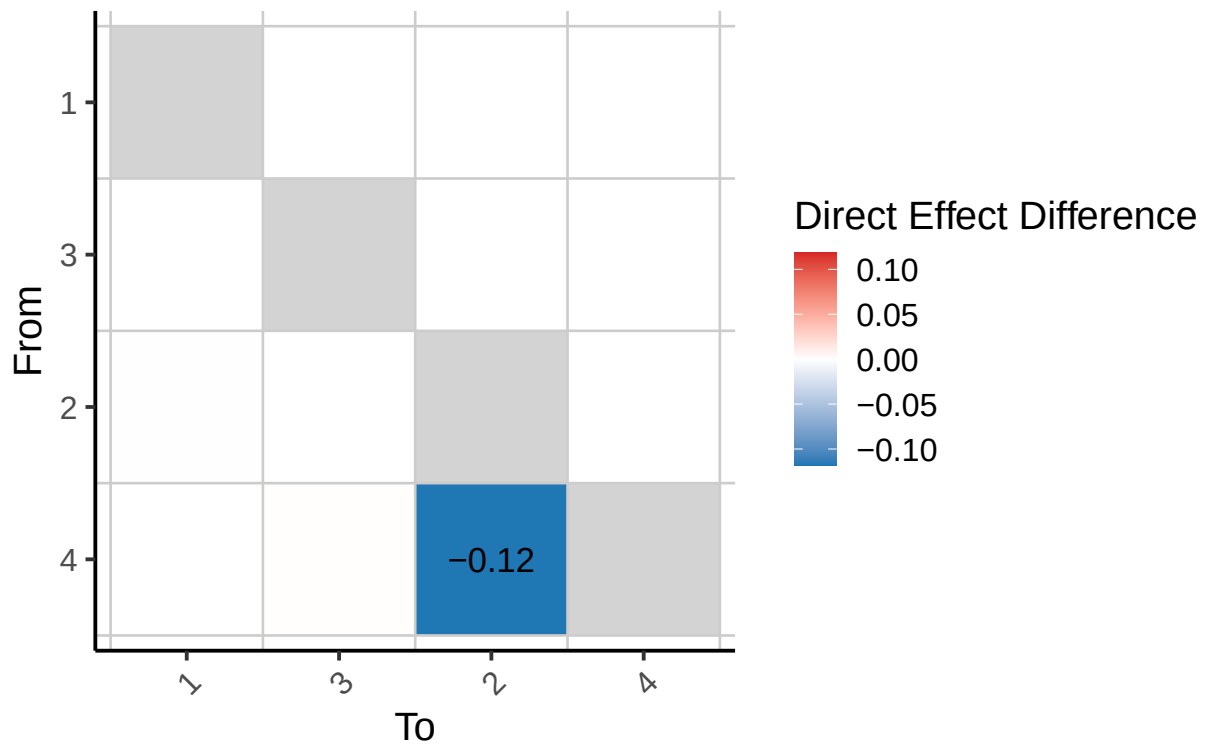
Graph Differences

Graph Differences - Beta difference

```
best_graph <- top_i_graph(mh_results)
next_graph <- top_i_graph(mh_results, i = 2)

plot_graph_differences(best_graph, next_graph, diff_type = "beta")
##> Warning: Removed 4 rows containing missing values or values outside the scale range
##> (`geom_text()`).
```

Effect Differences Between Graphs (Direct Effect)



```
# Adjacency (edge presence) differences  
plot_graph_differences(best_graph, next_graph, diff_type = "adj")
```

Edge Differences Between Graphs

